Written Assignment

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TYPE OF ASSIGNMENT:

☐ Pre-Module

☒ Post-Module

☐ Other

Part A — Functional Requirements Document

This document specifies a quantitative investment thesis and the application that implements it. Everything below was built and is running: the dashboard is published on GitHub Pages, the repository is public, and every figure in this document is read from the application's committed data files rather than typed in.

Live application: https://jakobhofer-kg16.github.io/equity-lens/#/portfolio

Repository: https://github.com/jakobhofer-kg16/equity-lens

1. Investment thesis

Name. “Cheap against peers, confirmed by the call.”

Buy S&P 500 companies that trade at a discount to their own industry peers on earnings and cash-flow multiples, whose most recent earnings call reads at least neutral and is not deteriorating, and whose share price is in a confirmed uptrend. Hold a diversified basket of 15–30 such names, weighted by a maximum-Sharpe optimisation within bands, and re-screen quarterly after each earnings season.

Why this and not something else. The thesis is not chosen from conviction; it is chosen from evidence produced by the same application. Before writing the thesis I backtested a five-factor fundamental score (growth, profitability, valuation versus peers, financial health, momentum) on 359 S&P 500 companies as of 2025-01-31 against their excess return over SPY in the following 252 trading days. The total score had no predictive power (rank correlation 0.00). One category did: **valuation versus industry peers, ρ = +0.14**, above the ±0.11 noise floor for that sample. Profitability was **negatively** related to subsequent returns (ρ = -0.13) — quality had already been priced. So the thesis keeps the one factor that showed signal, drops the one that hurt, and adds two things the score did not have: a text signal from the earnings call and a technical entry filter.

What the text signal adds. A low multiple is either an opportunity or a warning. The earnings call is where management has to defend the quarter to analysts. Scoring the call with a finance-specific word list (positive/negative polarity, hedging rate, legal vocabulary) and requiring the latest call to be at least neutral and not worse than the prior one is a cheap way of separating “cheap because ignored” from “cheap because in trouble”.

What the technical filter adds. Value stocks can stay cheap for years. The golden-cross and RSI rules taught in the module are used strictly as an entry timing filter: a name enters only when its 50-day average is above its 200-day average, the price is above the 200-day average, and RSI(14) is below 70. The filter does not select; it gates.

2. Equity universe

Screening universe. The 383 S&P 500 companies that have earnings-call transcripts in the course archive (kwartler/vienna-genai-finance-course). This is the widest set for which the text signal can be computed without a paid transcript API. 372 of them have Finnhub fundamentals.

Core universe (10 stocks). The ten highest-ranked names on the composite score, shown here with the figures that put them there. They are the reference set the functional requirements below are written against; the portfolio in Part B widens the selection to all names that pass the rules.

Ticker	Company	Industry	P/E vs peer	EV/EBITDA vs peer	Call tone (Δ)	RSI
GEN	Gen Digital Inc	Technology	17.7 / 27.7	11.3 / 22.5	0.93 (+0.06)	67
ALL	Allstate Corp	Insurance	5.0 / 13.1	4.3 / 10.3	0.71 (+0.11)	55
INCY	Incyte Corp	Biotechnology	16.1 / 20.3	12.2 / 22.0	0.87 (+0.02)	59
DAL	Delta Air Lines Inc	Airlines	13.1 / 23.0	7.6 / 15.7	0.80 (+0.04)	37
NVDA	NVIDIA Corp	Semiconductors	28.0 / 36.1	26.9 / 34.3	0.90 (+0.09)	59
CI	Cigna Group	Health Care	11.6 / 26.1	8.6 / 16.9	0.82 (-0.01)	56
FRT	Federal Realty Investment Tr	Real Estate	23.5 / 36.1	14.5 / 19.4	0.87 (+0.03)	42
AES	AES Corp	Utilities	5.6 / 21.0	11.3 / 13.3	0.80 (-0.00)	55
SPG	Simon Property Group Inc	Real Estate	14.8 / 36.1	19.4 / 19.4	0.81 (+0.03)	35
BBY	Best Buy Co Inc	Retail	14.6 / 20.3	9.1 / 13.4	0.77 (+0.17)	56

3. Selection criteria

3.1 Fundamental rules (the thesis)

– **Cheaper than peers.** Trailing P/E below the median of the company's Finnhub industry within the universe, or EV/EBITDA below the industry median. Both multiples must be positive (loss-making companies are excluded). Where an industry has fewer than four members the universe median is used.

– **Profitable.** Trailing operating margin above zero.

3.2 Text signal (earnings-call tone)

– **Tone level.** Polarity of the latest call — (positive − negative) ÷ (positive + negative) words over a finance lexicon — of at least 0.20. Management and analysts are separated by employer, not job title; operator lines are dropped; boilerplate (safe-harbour language, welcome) is excluded from highlights.

– **Not deteriorating.** Polarity change versus the prior call no worse than −0.10.

– **Not evasive.** Hedging words (may, could, uncertain, …) below 1.5% of all words.

3.3 Technical rules (entry filter, as taught)

– Golden cross: 50-day simple moving average above the 200-day.

– Price above the 200-day average.

– RSI(14) below 70 — not overbought at entry.

3.4 Ranking

Names that pass all rules are ranked by a composite: 50% average valuation discount to peers, 30% call-tone level, 20% call-tone change, each as a percentile rank. The industry cap (at most four per Finnhub industry) is applied on the way in.

4. Data sources

Source	Used for	Access	Limit that shaped the design
Finnhub (finnhub.io)	Company profile, 133 TTM metrics incl. P/E, EV/EBITDA, margins; industry; peers; quotes	Free API key, entered at run time	60 requests/minute; price history, price targets and transcripts are premium
Twelve Data	Daily OHLCV for the technical rules, the two-year return series for optimisation, and SPY	Free API key	8 requests/minute, 800/day; some symbols not on the free plan
Course transcript archive	Earnings-call transcripts (2,899 calls, 383 companies) for the text signal	Public GitHub, no key, CORS-enabled	Coverage stops at the archive's last quarter
OpenRouter	Executive commentary and thesis drafting (google/gemini-2.5-flash)	Personal API key, optional	Output is labelled AI-generated; never used for selection
Alpha Vantage	Consensus price target only	Free key, optional	25 requests/day; one request per company

Keys are entered in the application and stored in the browser only. The repository and the deployed bundle contain no key; this was verified with a search of the full Git history and of the published JavaScript.

5. Portfolio construction rules

– Hold every name that passes the rules, capped at 30 and floored at 15. Current count: 16.

– Weights from the maximum-Sharpe-ratio optimisation (the method chosen), long-only, between 2% and 10% per name. Minimum-variance and equal-weight are computed on the same holdings for comparison.

– Inputs: 504 trading days of daily log returns ending 2026-09-03; annualised mean and covariance; risk-free rate 4%.

– Re-screen quarterly after earnings season; exit a name when it breaks the 200-day average or its call tone falls below zero.

6. Application behaviour (functional requirements)

ID	Requirement	Acceptance criterion
FR-1	On load, fetch a live quote for every holding from Finnhub	All holdings show a price and day change within 10 seconds; failures are shown as “stale” with the last close, never hidden
FR-2	Display optimised weights and the equivalent $1M allocation	Weights sum to 100%; the user can switch between max-Sharpe, min-variance and equal weight
FR-3	Show a trading-signal summary per holding	Golden/death cross, RSI(14), MACD histogram sign and call tone with change are shown, with the as-of date
FR-4	Render an LLM executive commentary on load	With an OpenRouter key: four paragraphs written from the live quotes and signals, labelled with the model. Without a key: a computed signal summary and no pre-written text
FR-5	Compare the weighting methods	Expected return, volatility and Sharpe for all three methods and SPY over the same window, labelled in-sample
FR-6	Link every holding to the full research dossier	Clicking a ticker opens the company view with score, chart, peers, analysts, news and call sentiment
FR-7	Show the model's own track record	The backtest section states the rank correlation, quintile returns and per-category signal; its headline sentence is derived from the numbers

FR-8	Handle keys safely	Keys live in localStorage only; the setup screen appears when no Finnhub key is present; a bad key degrades one section, not the page
FR-9	Respect provider limits	Finnhub burst limit handled with one retry; Twelve Data used only in the offline build; nothing is fetched twice within six hours
FR-10	Publish via GitHub Pages	Every push to main builds and deploys; the URL is public

Non-functional

– Transparency: every number that is derived, reconstructed or model-generated is labelled as such next to where it appears.

– No invented data: without a key the application shows a setup screen, not sample companies.

– Tests: 41 unit tests over the scoring, sentiment, indicator and provider-parsing modules, including the invariant that analyst opinion changes the score by exactly zero.

Part B — Portfolio construction

Selection was run on 2026-09-04 with the rules in Part A. The funnel, the resulting holdings and the optimisation are below; the same numbers drive the dashboard.

1. Selection funnel

Step	Companies
S&P 500 companies with an earnings-call transcript in the course archive	383
… with Finnhub fundamentals	372
… cheaper than industry peers, profitable, acceptable call tone	171
… after the industry cap (max 4 per industry), priced	40
… on a golden cross, above the 200-day average, RSI < 70	16
Portfolio (top by composite rank)	16

The largest single reason for exclusion was not being cheaper than peers (128 companies), followed by loss-making businesses (40) and a deteriorating call (29). The technical filter then removed 24 of 40 priced candidates — cheap names that were still in a downtrend.

2. Holdings (16) and weights

Ticker	Company	Industry	Max Sharpe	Min var	P/E vs peer	Tone (Δ)	RSI
ALL	Allstate Corp	Insurance	10.0%	10.0%	5.0 / 13.1	0.71 (+0.11)	55
INCY	Incyte Corp	Biotechnology	10.0%	10.0%	16.1 / 20.3	0.87 (+0.02)	59
DAL	Delta Air Lines Inc	Airlines	10.0%	2.0%	13.1 / 23.0	0.80 (+0.04)	37
NVDA	NVIDIA Corp	Semiconductors	10.0%	8.0%	28.0 / 36.1	0.90 (+0.09)	59
SPG	Simon Property Group Inc	Real Estate	10.0%	10.0%	14.8 / 36.1	0.81 (+0.03)	35
SOLV	Solventum Corp	Health Care	10.0%	2.0%	10.3 / 26.1	0.67 (+0.06)	68
DVA	DaVita Inc	Health Care	10.0%	7.4%	13.7 / 26.1	0.59 (+0.22)	41
EXPE	Expedia Group Inc	Hotels, Restaurants &	10.0%	2.0%	18.0 / 19.9	0.91 (+0.04)	46
GEN	Gen Digital Inc	Technology	4.9%	8.7%	17.7 / 27.7	0.93 (+0.06)	67
OXY	Occidental Petroleum Corp	Energy	3.1%	10.0%	8.4 / 16.5	0.66 (+0.04)	60
CI	Cigna Group	Health Care	2.0%	10.0%	11.6 / 26.1	0.82 (-0.01)	56
FRT	Federal Realty Investment	Real Estate	2.0%	10.0%	23.5 / 36.1	0.87 (+0.03)	42
AES	AES Corp	Utilities	2.0%	3.9%	5.6 / 21.0	0.80 (-0.00)	55
BBY	Best Buy Co Inc	Retail	2.0%	2.0%	14.6 / 20.3	0.77 (+0.17)	56
EMN	Eastman Chemical Co	Chemicals	2.0%	2.0%	18.4 / 47.6	0.56 (+0.33)	45
DVN	Devon Energy Corp	Energy	2.0%	2.0%	16.4 / 16.5	0.88 (+0.07)	62

12 industries are represented; the largest weight is 10% and the top five names carry 50% of the max-Sharpe portfolio.

3. Optimisation results (in-sample)

Method	Expected return	Volatility	Sharpe
Maximum Sharpe (chosen)	19.1%	19.0%	0.79
Minimum variance	10.5%	16.0%	0.41
Equal weight	10.4%	18.1%	0.35
SPY, same window	15.8%	16.5%	0.72

The maximum-Sharpe weights are chosen for the mandate. Their expected return of 19.1% is a two-year historical mean — the weakest input in any mean-variance optimisation — which is exactly why the minimum-variance portfolio (which does not use expected returns) and equal weight are shown beside it. The 2%–10% band was added deliberately: without a floor the optimiser put ten names at the cap and nine at zero, which is a bet on last year's winners, not a diversified expression of the thesis.

Read the Sharpe ratios as a description of the fit, not a forecast: the weights were chosen on the same window they are measured on.

Part C — Executive Summary

Thesis. Buy S&P 500 companies that are cheaper than their own industry peers and whose latest earnings call confirms the business is not deteriorating; enter only in an uptrend. 16 names, weighted by maximum Sharpe ratio within 2%–10% bands, re-screened quarterly. Mandate: $1M.

Supporting logic. A backtest of a five-factor fundamental score on 359 S&P 500 companies as of 2025-01-31 found no power in the total score to rank the next year's excess returns (ρ 0.00), but a signal in valuation against industry peers (ρ +0.14) and a negative one in profitability (ρ -0.13). The strategy keeps the factor that worked and adds two checks the score lacked: earnings-call tone as a text signal against value traps, and the golden-cross/RSI rules as an entry filter against buying too early.

Data sources. Finnhub (fundamentals, industry, peers, live quotes; free, 60/min), Twelve Data (daily prices, free), the course transcript archive (2,899 earnings calls, no key), OpenRouter for commentary. Keys stay in the browser; none is in the repository or the published bundle.

Portfolio. 16 holdings across 12 industries; largest positions ALL 10%, INCY 10%, DAL 10%. In-sample: expected return 19.1%, volatility 19.0%, Sharpe 0.79 against SPY 0.72. Live at https://jakobhofer-kg16.github.io/equity-lens/#/portfolio.

Known weaknesses. (1) One backtest anchor is one draw. (2) Expected returns are two-year historical means; the max-Sharpe weights inherit their noise, which is why min-variance and equal weight are shown beside them. (3) The tone score is a word count and cannot read sarcasm or negation; four calls per company is a short history. (4) Small industries fall back to the universe median. (5) In-sample statistics describe the fit, not the future.

Ideas to strengthen. A second and third backtest anchor before trusting the valuation signal; shrunk or Black-Litterman expected returns; keys behind a serverless proxy plus SEC EDGAR fundamentals; the call-tone model checked against FinBERT; a turnover and cost model for the quarterly rebalance; and two quarters of paper trading before allocating.